

ASHUTOSH TIWARI

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WORK EXPERIENCE

Senior Machine Learning Engineer (ML Platform & Frameworks)

Jul. 2024 – Present

ADOBE FIREFLY

GENERATIVE AI, COMPUTER VISION, LARGE LANGUAGE MODELS

- Lead fault-tolerant distributed inference and training infrastructure for the org's ML platform: GPU pipelines on Kubernetes engineered to survive node loss and partial failure, from hyperscaled daily EMR ingestion through production model inference.
- **Distributed Inference Frameworks:** Designed and led engineering for the org's online and offline inference systems powering experimentation and production at scale, reaching 1000s×GPUs per model job across many model queues in one parallel wave.
 - Built online inference on vLLM + Ray for rapid experimentation with LLMs such as Falcon-40B and Llama 2 70B.
 - Built offline inference on Ray Data + PyTorch Lightning; a module benchmarked on that engine at 1.09M rows, 8×A100, 16 fractional-GPU actors. Introduced torch.compile(mode="max-autotune") in production: compiler-generated Triton kernels, fusion, CUDA-graph capture.
 - Modeled pipelines as horizontally-scaling plugins with no pod-to-pod communication, KEDA-autoscaled on SQS queue depth; batched sends retry on exponential backoff with full jitter, and undeliverable work drains to a DLQ.
- **Foundation Model Training Framework:** Leading development of a PyTorch-based training framework adopted across the org for training generative models on billions of images and videos.
 - Designed a plugin-based strategy pattern over FSDP/FSDP2 to make distributed training strategies pluggable.
 - Integrated Flash Attention 2/3, context parallelism, and distributed attention for DiT text-to-image and text-to-video training.
 - Grew the engineering bench: mentored engineers onto the framework, authored its onboarding guides, and reviewed their integrations.
- **Build & Deployment System:** Built a CLI-driven build system (BuildKit images on KEDA-scaled Kubernetes jobs, pushed to ECR), fronted by a Rust control plane and an MCP interface for AI-agent-driven builds, deployments, and inference.
- **Enrichment Service:** Orchestration over the inference framework; ran fire-and-forget pipelines behind a concurrency-safe S3 checkpoint cache, lease/TTL heartbeats and a circuit breaker.
- **Training-Data Governance:** Solely authored the org's fault-tolerant training-data lineage subsystem in Rust: idempotent claim/receipt registration, immutable S3 evidence layouts, and manifest-after-data publication.
- **Data Quality Framework:** Wrote the org's first data quality framework on Cerberus, used across enrichment pipelines to validate the correctness of feature-generation outputs across a Billion asset catalog growing 35–38M images/month.

Software Dev Engineer II (ML Platform)

Jan. 2019 – Jul. 2021

SWIGGY

DATA SCIENCE PLATFORM, TIME SERIES FORECASTING

- **Online Inference Platform:** Core contributor to the platform that served real-time model inference to 18M+ monthly transacting users across food delivery and quick commerce, autoscaled on live load metrics.
 - Engineered the dynamic request-routing layer that matched incoming API calls against geo-tags, traffic profile and service metadata, then dispatched them to model clusters partitioned by region and load domain.
 - Implemented routing and dispatch on an Akka-style actor framework: resilient asynchronous invocation, fault isolation across clusters, and backpressure-aware scheduling that held sustained low P99 under high QPS.
- **Feature Store:** Built and maintained the feature store and its pipelines, serving on-demand features to production ML models at 4Bn rows and 10K QPS, with multichannel ingestion via Spark, Flink, and user files.
- **Forecasting & Correlation Platform:** Founding member of the time-series forecasting platform adopted by teams across the org; its forecasts drove critical scaling decisions in real time.
- **DAQ:** Led DAQ, a large-scale API scraping tool collecting 15M rows daily for analysis and model training.

Software Development Engineer (Search Relevance)

Sep. 2017 – Jan. 2019

FLIPKART (a Walmart company)

SEARCH RELEVANCE, QUERY INTENT, NLP

- **Search Intent Models:** Built and improved CRF and neural-network intent models powering search and discovery for millions every day; drove the error-class analysis, designed the fixes and shipped them.
- **Tail Query Classifier:** Shipped a FastText-based query-store classifier that predicts the category of a tail query.
- **ML Workflow Automation:** Built the first workflow at Flipkart to retire manual deploys, automating training and auto-deployment of search models—written first in Luigi and later migrated to Airflow.
 - Wrote a generic Airflow framework that builds DAGs at runtime per ML model and orchestrates the train→deploy flow, gating every auto-deploy on data and model validations.
- **Large-Scale Data Pipelines:** Implemented Cascading/HDFS pipelines over 4Bn+ user-event datapoints, extracting and transforming them into the training data the search models learned from.

Software Development Engineer

Sep. 2016 – Sep. 2017

GROUPON

BACKEND ENGINEERING, RECOMMENDATION SYSTEMS

- Customer segmentation and deal-recommendation ML (NSVD unsupervised collaborative filtering on Neo4j); Cyclops customer-service platform live in every Groupon country.

Software Engineer

Sep. 2015 – Aug. 2016

NETSPEED SYSTEMS (Acquired by Intel)

GRAPH ALGORITHMS, NETWORK ON CHIP

- C++ graph algorithms for Network-on-Chip interconnects: Virtual Channel Arbitration, Polarity-based Arbitration, Multi-Cast Filtering, Structural Latency Breakdown.

PUBLICATIONS

Accepted at [NetSci 2023](#) (Poster Presentation) and [IC2S2 2023](#) (Parallel Talk) as **first author**

- [1] **Ashutosh Tiwari**, Prof. Sadamori Kojaku, Prof. Yong-Yeol Ahn, “Biased Contrastive Learning debiases Graph Neural Networks,” *International Conference on Network Science (NetSci)*, 2023. [Poster](#)

A non-parametric contrastive framework for fairness-aware graph representation learning: it debiases GNN embeddings with respect to sensitive node attributes and structural homophily for more organic recommendations.

EDUCATION

Indiana University Bloomington

Master of Science in Computational Data Science

Aug. 2021 – May 2023

National Institute of Technology, Patna

Bachelor of Science in Computer Science

Aug. 2011 – May 2015

RESEARCH EXPERIENCE / INDEPENDENT STUDY

- Debiasing independent study at [IUNI](#) with Profs. YY Ahn & S Kojaku — novel training methods for fairness-aware graph recommendation from biased graphs (the line of work behind the NetSci/IC2S2 publication).
- Paid Research Assistant on “**User Intent as a Network**” with Profs. YY Ahn, P Kantak & FB Yara, funded by the Kelly School of Business — quantifying user intent through network structure.
- [NLP Lab](#)@IUB with Prof. D Cavar on [TieML](#) and event-timeline modelling using fine-tuned large language models.

TEACHING EXPERIENCE

- Teaching Assistant for [Network Science](#) (INFO-I 606) with Prof. YY Ahn Spring 2023
- Teaching Assistant for [Machine Learning](#) (CSCI-B 555) with Prof. R Khardon Fall 2022
- Teaching Assistant for [Network Science](#) (INFO-I 606) with Prof. YY Ahn Spring 2022

SELECTED PROJECTS

rollout | *Rust, Distributed Training & Inference, RL for LLMs, vLLM, PyO3, Postgres*

github.com/thunderock/rollout

- Multi-node reinforcement-learning framework for LLMs in Rust (18 crates, extensive test coverage, CI): PPO/GRPO/DPO/SFT/RM algorithms, vLLM-backed batch + online inference, actor/learner split with work-stealing, training-state and CRIU process snapshots, PyO3 in-process plugins.

graph_ml | *Python, PyTorch Geometric, C++/Cython*

github.com/thunderock/graph_ml

- sklearn-style Graph-ML library on PyTorch/PyTorch Geometric with optimized C++/Cython random-walk kernels; CI-tested.

bias_manifold | *PyTorch, Graph ML, Fairness*

github.com/thunderock/bias_manifold

- Studies the structure and temporal evolution of bias manifolds across datasets — the fairness-aware graph-representation line of work behind the NetSci/IC2S2 publication.

BiasNet | *Reinforcement Learning, PyTorch, Actor-Critic*

github.com/thunderock/BiasNet

- Deep on-policy actor-critic RL agent that learns to play Street Fighter II from frame-difference observations (induced relational bias) over a large action space.

DeepFoodie | *Self-Supervised Learning, Embeddings*

github.com/thunderock/DeepFoodie

- Self-supervised clustering of food items using learned ingredient embeddings.

TECHNICAL SKILLS

Languages & Inference: Rust, C++, Python, PyO3, vLLM, ONNX Runtime, Triton, torch.compile, CUDA graphs, A100, H100

Training & Distributed: PyTorch, PyTorch Lightning, FSDP, Flash Attention, NCCL, Ray

RL & Alignment: PPO, GRPO, DPO, SFT, RLHF / reward modeling, actor-critic

Platform & Cloud: Kubernetes, KEDA, Docker / BuildKit, AWS (EMR, ECR, S3, SQS), Spark, Flink, Iceberg, Postgres

Concepts: tensor parallelism, pipeline parallelism, Mixture of Experts (MoE) parallelism, KV-cache management, continuous batching

ML & Research: Graph Neural Networks, Contrastive Learning, Recommendation Systems, PyTorch Geometric, TensorFlow, scikit-learn, Pandas, NumPy